

Vector spaces

 $(K, +_K, \cdot_K)$ field.

$\forall$ a space s.t. $(\forall) v \in V, (\forall) \alpha \in K : \alpha \cdot v \Rightarrow$
(vector) (scalar)

=> V vector space if: 1) $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$

$$2) \alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2$$

$$3) (\alpha \cdot \beta) \cdot v = \alpha \cdot (\beta \cdot v)$$

4) $1 \cdot v = v$

↓
id. elem. im $(K, \cdot_K)$

$$(\forall) \alpha, \beta \in K, (\forall) v, v_1, v_2 \in V.$$

$S \subseteq V$ vector space.

S subspace of V if: $\begin{cases} S \neq \emptyset \\ (\forall) \alpha, \beta \in K, (\forall) v_1, v_2 \in S, \alpha \cdot v_1 + \beta \cdot v_2 \in S. \end{cases}$

① K field.

(v) $k \in K$, (vi) $f \in K[x]$, $f = a_0 + a_1x + \dots + a_m \cdot x^m$

$$k.f = k.a_0 + (k.a_1)x + \dots + (k.a_n)x^n$$

$K[x]$ vector space?

1) (iv) $k_1, k_2 \in K$ scalars, (v) $f \in K[x]$ vector: $(k_1 + k_2)f \stackrel{?}{=} \underbrace{k_1}_{K_1} f + \underbrace{k_2}_{K_2} f$

$$H_1 = (k_1, k_2) (a_0 + a_1 x + \dots + a_n x^n)$$

$$= (k_1 + k_2)a_0 + (k_1 + k_2)a_1x + \dots + (k_1 + k_2)a_n \cdot x^n$$

$$= \underline{k_1} \cdot a_0 + k_2 \cdot a_0 + \underline{k_1} \cdot a_1 \cdot x + k_2 \cdot a_1 \cdot x + \dots + \underline{k_1} \cdot a_m \cdot x^m + k_2 \cdot a_m \cdot x^m$$

$$= k_1(a_0 + a_1x + \dots + a_mx^m) + k_2(a_0 + a_1x + \dots + a_mx^m)$$

$$= k_1 \cdot f + k_2 \cdot f = m_2$$

$\Rightarrow$ distributivity between the elements of the same field.

2) (v) $k \in K$, (vi) $f, g \in K[x]$, $g = b_0 + b_1x + \dots + b_m \cdot x^m$: $k(f+g) \stackrel{?}{=} k \cdot f + k \cdot g$
 $\mu_1 \quad \mu_2$
 Suppose $\text{degree}(f) > \text{degree}(g)$
 $(n > m)$

$$f = a_0 + a_1x + \dots + a_m \cdot x^m + \dots + a_n \cdot x^n$$

$$g = b_0 + b_1x + \dots + b_m \cdot x^m$$

$$\begin{aligned} \mu_1 &= k(f+g) = k(a_0 + a_1x + \dots + a_m \cdot x^m + b_0 + b_1x + \dots + b_m \cdot x^m) \\ &= k \cdot a_0 + k \cdot a_1x + \dots + k \cdot a_m \cdot x^m + \dots + k \cdot b_0 + k \cdot b_1x + \dots + k \cdot b_m \cdot x^m \\ &= k(a_0 + b_0) + k(a_1 + b_1)x + \dots + k(a_m + b_m) \cdot x^m + \\ &\quad + k \cdot a_{m+1} \cdot x^{m+1} + \dots + k \cdot a_n \cdot x^n \\ &= k a_0 + k a_1x + \dots + k a_m x^m + k b_0 + k b_1x + \dots + k b_m x^m \\ &= k \cdot f + k \cdot g = \mu_2 \end{aligned}$$

3) (v) $k_1, k_2 \in K$, (vi) $f \in K[x]$: $(k_1 \cdot k_2) \cdot f \stackrel{?}{=} k_1 \cdot (k_2 \cdot f)$
 $\mu_1 \quad \mu_2$

$$\begin{aligned} \mu_1 &= (k_1 \cdot k_2) f = (k_1 \cdot k_2) \cdot a_0 + (k_1 \cdot k_2) \cdot a_1 \cdot x + \dots + (k_1 \cdot k_2) \cdot a_m \cdot x^m \\ &= k_1 \cdot (k_2 \cdot a_0) + k_1 \cdot (k_2 \cdot a_1) \cdot x + \dots + k_1 \cdot (k_2 \cdot a_m) \cdot x^m \\ &= k_1 (k_2 \cdot a_0 + k_2 \cdot a_1 x + \dots + k_2 \cdot a_m x^m) \\ &= k_1 \cdot (k_2 \cdot f) = \mu_2 \end{aligned}$$

4) (v) $f \in K[x]$, e id. elem. of $(K, \cdot)$: $e \cdot f \stackrel{?}{=} f$
 $\mu_1 \quad \mu_2$

$$\begin{aligned} \mu_1 &= e \cdot f = e(a_0 + a_1x + \dots + a_m \cdot x^m) \\ &= e \cdot a_0 + e \cdot a_1x + \dots + e \cdot a_m \cdot x^m \\ &= a_0 + a_1x + \dots + a_m \cdot x^m \\ &= f = \mu_2. \end{aligned}$$

(1), (2), (3), (4) $\Rightarrow K[x]$ vector space.

④ $V = \{x \in \mathbb{R} \mid x > 0\}$

$x \cdot y = xy$, (v) $k \in \mathbb{R}$, (vi) $x, y \in V$
 $k \cdot x = x^k$ (scalar) (vectors)

V is a vector space?

V is an $\mathbb{R}$ -vector space:

$$1) (\forall) k_1, k_2 \in \mathbb{R}, (\forall) x \in V: \underbrace{(k_1 + k_2)^T x}_{M_1} \stackrel{?}{=} \underbrace{k_1^T x}_{M_1} + \underbrace{k_2^T x}_{M_2}$$

$$M_1 = x^{k_1 + k_2} = x^{k_1 \cdot k_2} = x^{k_1} \cdot x^{k_2} = k_1^T x + k_2^T x = M_2.$$

$$2) (\forall) k \in \mathbb{R}, (\forall) x, y \in V: \underbrace{k^T (x + y)}_{M_1} \stackrel{?}{=} \underbrace{k^T x}_{M_1} + \underbrace{k^T y}_{M_2}$$

$$M_1 = k^T x + y = x y^k = x^k \cdot y^k = k^T x + k^T y$$

$$3) (\forall) k_1, k_2 \in \mathbb{R}, (\forall) x \in V: \underbrace{(k_1 \cdot k_2)^T x}_{M_1} \stackrel{?}{=} \underbrace{k_1^T (k_2^T x)}_{M_2}$$

$$M_1 = x^{k_1 \cdot k_2} = (x^{k_1})^{k_2} = (x^{k_2})^{k_1} = (k_2^T x)^T k_1 = M_2.$$

$$4) (\forall) x \in V, e = \text{id. elem.} = 1: \underbrace{1^T x}_{M_1} \stackrel{?}{=} x$$

$$1^T x = x^1 = x$$

⑤ K field. $V = K \times K = K^2$; V is a vector space? $\left| \begin{array}{l} (\forall) x_1, y_1, x_2, y_2 \in V. \\ (\forall) k \in K. \end{array} \right.$

$$a) (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$k(x_1, y_1) = (k \cdot x_1, k \cdot y_1)$$

$$(\forall) k_1, k_2 \in K, (\forall) (x, y) \in V: \underbrace{(k_1 + k_2)(x, y)}_{M_1} \stackrel{?}{=} \underbrace{k_1(x, y)}_{M_1} + \underbrace{k_2(x, y)}_{M_2}$$

$$M_2 = k_1(x, y) + k_2(x, y) = (k_1 x, k_1 y) + (k_2 x, k_2 y) =$$

$$= (k_1 x + k_2 x, k_1 y + k_2 y)$$

$$= ((k_1 + k_2)x, (k_1 + k_2)y) \neq M_1.$$

$\Rightarrow$ not a vector space.

$$b) (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$k \cdot (x_1, y_1) = (k \cdot x_1, y_1)$$

$$1) (\forall) k_1, k_2 \in K, (\forall) (x, y) \in V: \underbrace{(k_1 + k_2)(x, y)}_{M_1} \stackrel{?}{=} \underbrace{k_1(x, y)}_{M_1} + \underbrace{k_2(x, y)}_{M_2}$$

$$M_1 = (k_1 + k_2)(x, y) = ((k_1 + k_2)x, y) = (k_1 x + k_2 x, y)$$

$$M_2 = (k_1 x, y) + (k_2 x, y) = (k_1 x + k_2 x, y + y)$$

$$= ((k_1 + k_2)x, 2y)$$

$M_1 \neq M_2 \Rightarrow$ not a vector space

⑦ Subspaces of $\mathbb{R}^3$?

a) $A = \{ (x, y, z) \in \mathbb{R}^3 \mid x=0 \}$

$A \neq \emptyset, (0, 1, 1) \in A \Rightarrow x=x'=0.$

$(\forall) \alpha_1, \alpha_2 \in \mathbb{R}, (\forall) a_1, a_2 \in A: \alpha_1 a_1 + \alpha_2 a_2 \stackrel{?}{\in} A$

$\alpha_1(x, y, z) + \alpha_2(x', y', z') \stackrel{?}{\in} A$

$(\alpha_1 x, \alpha_1 y, \alpha_1 z) + (\alpha_2 x', \alpha_2 y', \alpha_2 z') \stackrel{?}{\in} A$

$(\underbrace{\alpha_1 x + \alpha_2 x'}_0, \alpha_1 y + \alpha_2 y', \alpha_1 z + \alpha_2 z') \stackrel{?}{\in} A$
 $\Rightarrow A$ subspace.

b) $B = \{ (x, y, z) \in \mathbb{R}^3 \mid x=0 \text{ or } z=0 \}$

counterexample: $\exists (0, 1, 1), (1, 1, 0) \in B \xrightarrow{1 \in \mathbb{R}} 1 \cdot (0, 1, 1) + 1 \cdot (1, 1, 0) = (1, 2, 1) \notin B$

$\Rightarrow B$ not a subspace.

c) $C = \{ (x, y, z) \in \mathbb{R}^3 \mid x \in \mathbb{Z} \}$

$\exists (2, 1, 1) \in C \xrightarrow{\sqrt{2} \in \mathbb{R}} (2\sqrt{2}, \sqrt{2}, \sqrt{2}) \notin C$

$\Rightarrow C$ not a subspace.

d) $D = \{ (x, y, z) \in \mathbb{R}^3 \mid x+y+z=0 \}$

$D \neq \emptyset, (1, 0, -1) \in D \Rightarrow x_1+y_1+z_1 = x_2+y_2+z_2 = 0.$

$(\forall) \alpha_1, \alpha_2 \in \mathbb{R} \xrightarrow{?} \alpha_1 a_1 + \alpha_2 a_2 \stackrel{?}{\in} D.$

$(\forall) a_1, a_2 \in D \searrow$

$\alpha_1(x_1, y_1, z_1) + \alpha_2(x_2, y_2, z_2) =$

$= (\alpha_1 x_1, \alpha_1 y_1, \alpha_1 z_1) + (\alpha_2 x_2, \alpha_2 y_2, \alpha_2 z_2)$

$= (\underbrace{\alpha_1 x_1 + \alpha_2 x_2}_x, \underbrace{\alpha_1 y_1 + \alpha_2 y_2}_y, \underbrace{\alpha_1 z_1 + \alpha_2 z_2}_z)$

$\left. \begin{array}{l} x_1 + y_1 + z_1 = 0 \\ x_2 + y_2 + z_2 = 0 \end{array} \right\} \Rightarrow x_1 + y_1 + z_1 + x_2 + y_2 + z_2 = 0$

$$x+y+z = \alpha_1 \underbrace{(x_1+y_1+z_1)}_0 + \alpha_2 \underbrace{(x_2+y_2+z_2)}_0 = 0.$$

$\Rightarrow D$ subspace.

$$e) E = \{(x, y, z) \in \mathbb{R}^3 \mid x+y+z=1\}$$

$$\left. \begin{array}{l} \exists (1, 0, 0) \in E \\ \sqrt{2} \in \mathbb{R} \end{array} \right\} \Rightarrow (\sqrt{2}, 0, 0) \notin E$$

$\Rightarrow E$ not a subspace.

$$f) F = \{(x, y, z) \in \mathbb{R}^3 \mid x=y=z\}$$

$$F \neq \emptyset, \exists (1, 1, 1) \in F.$$

$$\left. \begin{array}{l} (x) \alpha_1, \alpha_2 \in \mathbb{R} \\ (y) a_1, a_2 \in F \end{array} \right\} \xrightarrow{?} \alpha_1 a_1 + \alpha_2 a_2 \xrightarrow{?} F$$

$$\alpha_1(x_1, y_1, z_1) + \alpha_2(x_2, y_2, z_2) \xrightarrow{?} F$$

$$(\alpha_1 x_1, \alpha_1 y_1, \alpha_1 z_1) + (\alpha_2 x_2, \alpha_2 y_2, \alpha_2 z_2) \xrightarrow{?} F$$

$$= (\alpha_1 x_1 + \alpha_2 x_2, \alpha_1 y_1 + \alpha_2 y_2, \alpha_1 z_1 + \alpha_2 z_2) \xrightarrow{?} F$$

$$= \left(\underbrace{\alpha_1 x_1 + \alpha_2 x_2}_x, \underbrace{\alpha_1 y_1 + \alpha_2 y_2}_y, \underbrace{\alpha_1 z_1 + \alpha_2 z_2}_z \right) \xrightarrow{?} F$$

$$x=y=z \Rightarrow F \text{ subspace.}$$

$$\textcircled{9} \quad a) K_m[x] = \{f \in K[x] \mid \deg(f) \leq m\} \quad m \in \mathbb{N}.$$

$$b) K'_m[x] = \{f \in K[x] \mid \deg(f) = m\}$$

Subspaces of ${}_K K[x]$ vector space?

$$b) f = x^m \in$$

$$g = -x^m \in K'_m[x]$$

$$f+g = x^m - x^m = 0 \Rightarrow \deg(0) = -\infty < m$$

$\Rightarrow K'_m[x]$ not a subspace.

! a) K field.

$$K_m[x] = \{f \in K[x] \mid \deg(f) \leq m\}$$

We know: $K[x]$ (all polynomials over K) is a K -vector space under usual $+$ and $\cdot$.

$W \subseteq V$ of a vector space V is a subspace $\Leftrightarrow \begin{cases} W \neq \emptyset \\ u+v \in W, (\forall) u, v \in W. \\ k \cdot u \in W, (\forall) k \in K \\ (\forall) u \in W. \end{cases}$

$W \neq \emptyset$

$0 \in K_m[x]$ because $\text{degree}(0) = -\infty < m \Rightarrow K_m[x] \neq \emptyset$ (1)

$u+v \in W, (\forall) u, v \in W$

$f, g \in K_m[x] \Rightarrow \text{degree}(f) \leq m$
 $\Rightarrow \text{degree}(g) \leq m$

Property: $\text{degree}(f+g) \leq \max(\text{degree } f, \text{degree } g) \leq m.$

$\Rightarrow f+g \in K_m[x].$ (2)

$k \cdot u \in W, (\forall) k \in K, (\forall) u \in W.$

$(\forall) k \in K, f \in K_m[x]: \text{degree}(k \cdot f) = \begin{cases} \text{degree } f, & k \neq 0 (\leq m \checkmark) \\ -\infty, & k = 0 (\leq m \checkmark) \end{cases}$

$\Rightarrow k \cdot f \in K_m[x]$ (3)

(1), (2), (3) $\Rightarrow K_m[x]$ subspace of $K[x]$.